CAITLYNN YEAR

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EDUCATION

Columbia University — New York, USA
Bachelor of Science in Mechanical Engineering, Minor in Computer Science — Expected May 2028
Relevant Coursework: Machine Design, Mechatronics, Continuous Control Systems, Thermodynamics, Mechanics, Electrical Engineering, Data Structures, Machining.

EXPERIENCE

Celestica — Richardson, TX
Mechanical Engineer Intern — June 2026 – Aug 2026

- Designed and validated a load-bearing hardware fixture in SolidWorks with GD&T drawings, structural FEA in ANSYS, and iterative prototyping, replacing an unsafe 110-lb lift, cutting install time 50% and eliminating recurring part damage
- Developed custom end-of-arm tooling, prototypes, and part-presentation fixtures eliminating manual handling while improving robotic placement repeatability by commissioning two 6-axis robotic arms for automated rack assembly
- Led root cause analysis of a coolant leak that failed a $5M server rack, performing pressure testing and subsystem validation to identify failure mechanisms and authored report to hyperscaler that updated SOPs and enabled shipment
- Created and deployed 4 custom assembly tools in Onshape increasing assembly output by 30% by eliminating scanning errors, protecting delicate fiber-optic cables from handling damage, and reducing operator labor from minutes to seconds
- Removed 4 material-delivery bottlenecks capping output from 3 to 8 racks/day by designing a kanban point-of-use system for 8 assembly stations, sizing bins by interpreting assembly BOMs and engineering drawings from the global design team
- Implemented leak-test layouts in AutoCAD reducing test time 25% by analyzing tool placement and operator workflows

Columbia University — New York, NY
Course Assistant – Industrial Engineering and Operations Research Department — Sept 2025 – Present

- Authored the master logistics and grading standard adopted by 3 course assistants for a 250-student graduate data analytics course, cutting new CA onboarding from days to a few hours and standardizing grading turnaround for all CAs

PROJECTS

Animatronic Facial Tracking — June 2026 – Aug 2026

- Developed a compact 3-DOF robotic eye mechanism using snap-fit assemblies, ball-joint linkages, and servo-driven actuation to achieve human-like gaze and blinking while supporting repeated assembly without fasteners
- Implemented real-time face tracking in OpenCV on a Raspberry Pi that continuously converted facial feature detection into closed-loop servo commands, enabling smooth autonomous gaze tracking of moving faces and finger gestures

Automated Thermal Management Enclosure — Apr 2026 – June 2026

- Built a closed-loop cooling system with IR temperature sensing and cooling fan control to hold an ESP32 below its 60°C throttling threshold under sustained load, cutting steady-state temperature 4°C and removing intermittent throttling entirely
- Optimized heatsink fin geometry and enclosure airflow with thermal FEA in ANSYS, benchmarked against hand-calculated convection estimates within 2°C to validate models, then generated the CAM program to machine the most optimal design

Reverse Engineering Consumer Electronics — Jan 2026 – Apr 2026

- Redesigned a failing smartwatch hinge, using worst-case tolerance stack-up analysis to trace the failure to fatigue cracking in the main joint that secured the wristband to the arm and iterating a fix validated across printed prototype cycles
- Reverse-engineered 4 consumer products into fully-toleranced CAD models with complete GD&T drawings in Siemens NX, validating against originals with 3D-printed prototypes that showcased wall-thickness variation between products

Robotics Club - Locomotives Subteam (Batteries Subdivision) — Sept 2025 – Dec 2025

- Designed and validated 3 LiPo battery enclosures in Fusion 360, cutting pack weight 10% against a mandated competition weight budget by switching to an interlocking TPU design while still preserving puncture safety for the cells

NASA Mission Concept Academy — June 2025 – Aug 2025

- Designed a lightweight, high-strength rover frame in Siemens NX to meet stringent mission science objectives, utilizing finite element analysis to optimize the structural-to-weight ratio ensuring interface compatibility with adjacent subsystems

SKILLS

Mechanical Design: SolidWorks, NX, Onshape, GD&T, Tolerance Analysis, DFM/DFA, ANSYS (Structural/Thermal)
Software & Embedded Systems: Arduino, ESP32, Raspberry Pi, Python, C++, Java, MATLAB, OpenCV, Git
Manufacturing: CAM, CNC, Lathe, Rapid Prototyping, Soldering, Lean, Six Sigma